

Freeze ablation – results

Three runs sharing the same setup (HCP-only, T=1200, temporal_kernel=20, base_lr=3e-4, effective batch 16, 20 epochs, DINOv2 ViT-S/14 reg4 ImageNet init). Only the parameter-freezing policy differs.

Architecture: what we added on top of DINOv2

We keep the DINOv2 SSL algorithm (student-teacher multi-crop self-distillation with iBOT masked-patch prediction) unchanged. Four fMRI-specific pieces are added on top of the official code:

1. **3D+1D patchify** (PatchEmbed3DPlus1D). Replaces DINOv2's 2D Conv2d patch embed. A hierarchical encoder built from Conv3Plus1d blocks (separable spatial-3D + temporal-1D conv, in the MovieGen-TAE style). Input (T, X, Y, Z) -> output token grid ($T_{\text{eff}} = T / \text{temporal_kernel}$, $N_{\text{spatial}} = N_x * N_y * N_z$). For HCP at T=1200 with temporal_kernel=20 and (45, 54, 45) voxels at patch_size=9: yields $60 * 150 = 9000$ patch tokens, plus 1 CLS + 4 register tokens = 9005 tokens fed to the ViT.
2. **Factorised positional embedding** (PositionEmbedding3D). Three small tables – pos_temporal (T_{eff}), pos_spatial (N_{spatial}), pos_cls (1) – broadcast and summed. $O(T_{\text{eff}} + N_{\text{spatial}})$ parameters instead of $O(T_{\text{eff}} * N_{\text{spatial}})$ for a flat 2D ViT pos table.
3. **3D multi-crop augmentation** (MultiCrop3D). For each scan, picks 2 global + 8 local random 3D spatial sub-volumes (scale $[0.32, 1.0]$ and $[0.05, 0.32]$ of the volume) and trilinearly resizes each back to (X, Y, Z). No photometric augmentation (color jitter, blur, solarize, ImageNet normalize) and no flip (preserves L/R brain asymmetry) – those don't apply to z-scored fMRI volumes.
4. **fMRI datasets** (HCPFullScanDataset, ADNIFullScanDataset, MixedFMRIDataset). Each scan loaded as (T, 1, X, Y, Z) with per-frame z-score. HCP scans are reconstructed from saved windowed .pt files; ADNI scans are slices of a shared 5D tensor.

Everything else (the SSL training loop, EMA teacher, DINO + iBOT + KoLeo losses, FSDP wrapping, sinkhorn-knopp centering, schedulers) is the official DINOv2 code, **unchanged**.

Strategies and trainable parameter counts

Trainable parameter counts come from the actual training logs (backbone trainable=... line printed by apply_freeze_policy). Head counts are reported by scripts/count_trainable_params.py. Heads (DINO-Head) are random-init and **always trainable**, regardless of the freeze policy.

Strategy	Backbone trained	Backbone trainable	Backbone frozen	Heads	TOTAL trainable
A – Full fine-tune	all blocks + patch_embed + norm	36.21 M	0	~6.55 M	~42.8 M
B – Freeze last 3	patch_embed + blocks.9-11 + norm	16.78 M	19.44 M	~6.55 M	~23.3 M
C – Freeze except input	patch_embed only	11.45 M	24.76 M	~6.55 M	~18.0 M

The backbone contains PatchEmbed3DPlus1D (~11.45 M – much bigger than a 2D ViT patch embed because of the temporal Conv1Plus3d blocks), 12 transformer blocks (~21.2 M total), and the dead-weight 2D pos_embed allocated by the ViT parent class (~3.46 M, unused in fMRI mode). The DINO head is a 3-layer MLP with 4096 prototypes (~6.55 M).

Best checkpoint per strategy (selected among available iters: 5999/8999 for A, 11999/17999 for B and C).

HCP probes

Label	Metric	Chance baseline	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.500	0.751	0.696	0.844
BrainVol	MAE	run script	116,179	117,744	99,455
GrayMatterVol	MAE	run script	34,762	34,815	29,274
Age	MAE	run script	3.044	3.056	2.931
FluidIntel	MAE	run script	4.047	4.035	4.003
ProcSpeed	MAE	run script	16.343	16.401	16.207
WorkingMem	MAE	run script	10.442	10.410	10.411

ADNI probes

Label	Metric	Chance baseline	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.500	0.653	0.696	0.813
CDR	AUC	0.500	0.542	0.522	0.532
Degradation1Y	AUC	0.500	0.581	0.543	0.559
Degradation2Y	AUC	0.500	0.533	0.456	0.533
Degradation3Y	AUC	0.500	0.528	0.468	0.544
MMSE	MAE	run script	2.478	2.426	2.750
Age	MAE	run script	5.113	4.993	5.703

Bold = best value in row. AUC: higher is better. MAE: lower is better. The “chance baseline” column shows a `DummyClassifier(strategy='stratified')` AUC (= 0.5 for any balanced binary task) or a `DummyRegressor(strategy='mean')` MAE – fill in from `scripts/compute_label_baselines.py` for the MAE rows.

Source data: `outputs/probes/probe_{adni,hcp}<run_name>_iter<iter>.json` – generated by `slurm_jobs/probe_{adni,hcp}.sh`, aggregated by `python3 scripts/summarize_probes.py`. Trainable-param counts: `python3 scripts/count_trainable_params.py`. Chance baselines: `python3 scripts/compute_label_baselines.py`.

Note on A: the original run (`dinov2_fmri_20260524_144327`) stopped at iter 10410 instead of completing the 20 epochs (19999 iters), so its results aren't strictly at the same training depth as B and C. Re-running A to completion (`fmri_vits_hcp_baseline.yaml`) for a fair comparison.